

Qns	Answer	Marks
8(a)(i)	In a polarised wave, the <u>oscillations</u> are <u>along one direction only</u> .	B1
	in a <u>single plane</u> that is <u>normal to the direction of energy transfer of the wave</u>	B1
8(a)(ii)	Light waves that have a constant phase difference.	B1
8(b)(i)	$\sin \theta = \frac{\lambda}{x}$ For small angles, $\sin \theta \approx \tan \theta$ $\frac{\lambda}{x} = \frac{\frac{1}{2}w}{D}$ $\frac{590 \times 10^{-9}}{x} = \frac{\frac{1}{2}(30.7 \times 10^{-3})}{2.6}$ $x = 1.0 \times 10^{-4} \text{ m}$	C1 A1
8(b)(ii)	$\theta > \frac{\lambda}{x} = \frac{590 \times 10^{-9}}{1.0 \times 10^{-4}}$ $\approx 0.0059 \text{ rad}$	C1 A1
8(b)(iii)	The <u>continuous central maximum</u> is replaced by an <u>interference pattern</u> of equally spaced <u>bright and dark fringes</u> within the single slit envelope.	B1 B1
8(b)(iv)	$\frac{I_{\text{single}}}{I_{\text{double}}} = \frac{kA^2}{k(2A)^2} = \frac{1}{4} = 0.25$	B1
8(b)(v)	1. Waves from both sources of light might be <u>polarized in perpendicular planes</u>. 2. Waves from both sources of light <u>do not have constant phase difference</u> OR are <u>not coherent</u> OR <u>have different wavelength/frequencies</u>. 3. Waves might have <u>different amplitudes</u>. (The resulting interference pattern will have less contrast due to incomplete cancellation at the minima.) Any two above.	B1 B1
8(c)(i)	Negative energy levels indicate that the atom is a bound system - electron cannot escape from the atom. OR Energy must be supplied to the system to bring the electron to infinity. OR The potential energy is taken to be 0 at infinity, and negative work needs to be done to move the electron from infinity to a point. Hence, the electric potential energy is negative.	B1
8(c)(ii)	10.38 eV	B1
8(c)(iii)(1)	Downward arrows All three arrows	B1 B1

8(c)(iii)(2)	Longest wavelength corresponds to the lowest transition energy: $-1.57 - (-5.74) = 4.17 \text{ eV}$ $\Delta E = \frac{hc}{\lambda}$ $4.17 \times 1.6 \times 10^{-19} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda}$ $\lambda = 2.98 \times 10^{-7} \text{ m} = 298 \text{ nm}$	B1 B1
8(c)(iv)	Since <u>the energy of the photon (8.9 eV) does not match the energy difference between any two of energy levels</u>, the electrons will not be excited. Hence, there is <u>no spectral lines</u>.	B1 B1

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9(a)(i)	Diffraction is the spreading of a wave into the geometric shadow when it	B1